MATH 401. Homework #3. Sadullaev L.

1.5. Problem 3.

Let $\mathcal{L}$ be the collection of all lines in the plane. Define a relation on $\mathcal{L}$ by saying that 2 lines are equivalent if and only if they are parallel or equal. Show that this is an equivalence relation on $\mathcal{L}$.

Solution.

Let $\mathcal{L}$ be the collection of all lines in $\mathbb{R}^2$. Let $\sim$ be a relation on $\mathcal{L}$ defined as

(1) $l \sim q \iff l$ and q have the same slop.

This (1) is an equivalence relation since:

(i) $l \sim l$ (slop of l & l are equal)

(ii) $l \sim q \Rightarrow$ slop of l = slop of $q = m \Rightarrow$ slope of q = slop of $l = m \Rightarrow q \sim l \Leftrightarrow$ $l \sim q \Rightarrow q \sim l$ holds.

(iii) $\ell \sim q$ (2), $q \sim p$ (3) $\Rightarrow$

$$\begin{cases} \text{slope of } \ell = \text{slope of } q = m \text{ by } (2) \\ \text{slope of } q = \text{slope of } p = m \text{ by } (3) \end{cases} \Rightarrow$$

slope of $\ell = m =$ slope of p (4) $\Rightarrow$

$\Rightarrow \ell \sim p$ (Transitivity holds as well)

So, $\sim$ is an equivalence relation. $\blacksquare$

1.5. Problem 6.

Let m be a natural number. Define a relation $\equiv$ on $\mathbb{Z}$ by

(1) $x \equiv y \iff (x-y)$ is divisible by m.

① Show that $\equiv$ is an equivalence relation.

② How many equivalence classes are there? What do they look like.

Solution:

To show $\equiv$ is an equivalence relation I'm going to show it satisfies all (i)-(iii) 3 properties of the definition of eq. relation:

(i) $x \equiv x \iff (x-x) = 0$ is divisible by m.

(i) holds $\forall x \in X = \mathbb{Z}$.

(ii) $x \equiv y \Rightarrow (x-y) \mid m = k, \; k \in \mathbb{Z}$.

$\Rightarrow$ Now let's see $\dfrac{(y-x)}{m} = \dfrac{-(x-y)}{m} = (-k) \in \mathbb{Z} \Rightarrow$

$y \equiv x$ holds, that is

$x \equiv y \Rightarrow y \equiv x$.

(iii) $x \equiv y \Rightarrow (x-y) \mid m = k, \; k \in \mathbb{Z} \; (2)$

$y \equiv z \Rightarrow (y-z) \mid m = p, \; p \in \mathbb{Z} \; (3)$

By (2) $\left.\begin{array}{l} x-y = k \cdot m \\ y-z = p \cdot m \end{array}\right\}$, $\Rightarrow$
By (3) +

$x-y+y-z = x-z = k \cdot m + p \cdot m \quad (4)$

$\Rightarrow x-z = (k+p) \cdot m \Rightarrow (x-z) \mid m = k+p$

for some $p, k \in \mathbb{Z} \Rightarrow$

$(x-z) \mid m \Rightarrow x \equiv z \Rightarrow$

$x \equiv y,\ y \equiv z \Longrightarrow x \equiv z$

Transivity holds as well, hence, $\equiv$ is an equivalence relation $\blacksquare$

② How many equivalence classes are there?

- For this question, intuitively, I was assuming for a relation there must be only one equivalence relation, since we showed that if $x \in [x]$, $x \in [y]$ for $x, y \in X$ s.t. $x \neq y$, then $[x] = [y]$ by subset inclusion way during the proof of Proposition 1.5.3.

- Also after definition 1.5.2. I encountred that there is a 1-to-1 correspondence between partitions (equivalence classes) & equivalence relation.

So, the only eq. class for $\equiv$ is

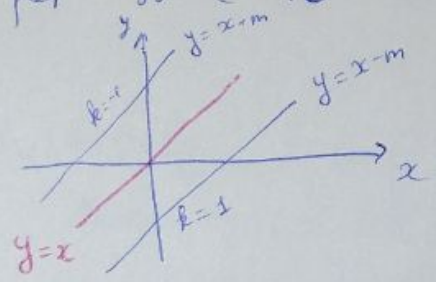
$$x \equiv y \iff (x-y) \mid m \implies$$

$$[x] = \{ y \in \mathbb{Z} \mid (x-y) \mid m \} \qquad (5)$$

How $[x]$ look like?

Set $\frac{x-y}{m} = k$ for some $k \in \mathbb{Z} \implies$

$$y = x - k \cdot m \qquad (6)$$



So, $$[x] = \underbrace{\{ y \in \mathbb{Z} \mid \frac{x-y}{m} = k \}}_{} \qquad (7)$$

collection of all lines that are obtained by transforming $y = x$ line by $(k \cdot m)$ units for some $k, m \in \mathbb{Z}$.

1.5. Problem 7.

Let $\mathcal{M}$ be the set of all $n \times n$ matrices. Define a relation (on) $\sim$ on $\mathcal{M}$ by

$A \sim B \iff \exists$ an invertible matrix $P \ni$

$$A = P^{-1} B P$$

Show $\sim$ is an eq. relation.

Solution:

[Preamble]

Similar technique with previous Pr. 6.

(i) $A \sim A$ since

$$A = P^{-1} A P \quad \{* P (l) \Rightarrow$$
$$PA = AP \quad \{\cdot P^{-1} (r) \Rightarrow$$
$$PAP^{-1} = A \Rightarrow$$
$$A \sim A.$$

for $\forall$ invertible P.

(ii) $A \sim B \Rightarrow$

$$A = P^{-1} B P \Rightarrow$$
$$\underbrace{P}_{S^{-1}} A \underbrace{P^{-1}}_{S} = B \quad [P^{-1} = S] \Rightarrow$$
$$S^{-1} A S = B \Rightarrow$$

$\Rightarrow B \sim A$

(iii) $A \sim B \Rightarrow A = P^{-1} B P$

$B \sim D \Rightarrow \boxed{B = S^{-1} D S} \Rightarrow$

$$A = P^{-1} B P = \underbrace{P^{-1} S^{-1}} D \underbrace{S P} = \underbrace{(SP)}_{R}^{-1} D \underbrace{(SP)}_{R}$$

$\Rightarrow A = R^{-1} D R \Rightarrow A \sim D$

So, $A \sim B$, $B \sim D \Rightarrow A \sim D$

$\blacksquare$

1.6. Problems.

Show that the set of all finite sequences $(q_1, q_2 \ldots q_k)$ $k \in \mathbb{N}$ of rational numbers is countable.

Solution:

To my understanding, $\{q_i\}$'s themselves are finite. So, I am following this thought.

Let $(q_1\ q_2 \dots q_k)$ $k \in \mathbb{N}$ be the set of all finite sequences. Also

Let $\{q_1\}, \{q_2\} \dots \{q_k\}$ be finite sequences of rational numbers. (1)

WTS: $\exists$ an ordered list for all elements of the set $(q_1\ q_2 \dots q_k)$.

! Since $\{q_i\}$'s are finite sequences, <u>they are sure countable</u> and have their own ordered fashion list maybe as follows:

$$\{q_i\} = \{q_{i1}, q_{i2}, q_{i3}, \dots q_{in}\} \qquad (2)$$

for some $n \in \mathbb{N}$ and $i = 1, 2, \dots k$.

Then, I can use the "index trick" used in Prop. 1.6.2 as follows

$$\{q_{\underset{2}{11}}, q_{\underset{3}{12}}, q_{\underset{3}{21}}, q_{\underset{4}{13}}, q_{\underset{4}{22}}, q_{\underset{4}{31}}, q_{\underset{5}{14}}, q_{\underset{5}{23}}, q_{\underset{5}{32}} \dots\} \qquad (3)$$

(3) is a listing of the set $(q_1 \dots q_k)$ s.t. the sum of indices increases as natural numbers.

Hence, we've found the ordered list we aimed to find. $\Rightarrow$

$(q_1, q_2 \dots q_k)$ is countable.

1.6. Problem 5.

Show that the set of all subsets of $\mathbb{N}$ is uncountable.

Solution:

Let $\mathcal{A} = \{A \mid A \subseteq \mathbb{N}\}$ (1)

Assume for contradiction that $\mathcal{A}$ is countable. $\Rightarrow$

$\exists$ a list to order **ALL** elements of $\mathcal{A}$ as follows

$$\mathcal{A} = \{ A_1, A_2, A_3 \dots \} \qquad (2)$$

Now let's write down what each subset A_i contains

$$A_1 = \{ a_{11}, a_{12}, a_{13}, a_{14} \dots \}$$
$$A_2 = \{ a_{21}, a_{22}, a_{23}, a_{24} \dots \} \qquad (3)$$
$$A_3 = \{ a_{31}, a_{32}, a_{33}, a_{34} \dots \}$$
$$\dots\dots\dots\dots$$

! We can write each $A_i \subseteq \mathbb{N}$, <u>since $\mathbb{N}$ is itself countable, so is A_i.</u> $\Rightarrow$ (3) ordered lists exist.

To get our contradiction, I introduce a new set $B = \{ b_1, b_2, b_3, b_4 \dots \}$ where the numbers b_i's are designed by

$$b_i = a_{ii} + 1 \qquad (4)$$

This set B with b_i elements has to be <u>different from the i-th subset A_i</u> on the list (3) by <u>at least one number b_i</u>

since the construction of b_i element of set B always differs by 1 from a_{ii} element of A_i subset. $\Rightarrow$

We could creat a new set B which is not in the family $\mathcal{A}$. $\Rightarrow$ $\mathcal{A}$ is uncountable. $\blacksquare$